Millikan's oil drop

Aim

To measure the charge of the electron

Analysis

We can calculate the values of the required constants using the measured values.

$$\rho = 963\ kg/m^3$$
$$\rho_0 = 1.8\ kg/m^3$$
$$\eta = 1.8363\ kg/ms$$
$$g = 9.78413\ m/s^2$$
$$C = 196.971$$
$$D = 8.7866 \times 10^{-9}$$
$$\zeta = 4.07 \times 10^{-8}$$

v_f	v_r	v_f	V	T	r	ne
19.28	21.73	1.04E-04	335	1.89	9.15E-07	8.50E-19
24.48	24.93	8.17E-05	350	1.98	8.08E-07	5.87E-19
21.43	40.31	9.33E-05	686	1.53	8.66E-07	2.85E-19
19.29	10.23	1.04E-04	510	2.89	9.15E-07	8.53E-19
11.86	12.44	1.69E-04	727	1.95	1.18E-06	8.63E-19

Table 1: Data calculated for dynamic methord

ne/min	n_{eff}	ne/n_{eff}
2.98	3	2.83E-19
2.06	2	2.94E-19
1.00	1	2.85E-19
2.99	3	2.84E-19
3.03	3	2.88E-19

The charge on a droplet is given by the formula

$$ne = \frac{Cr^3}{V}\left(1 + \frac{t_f}{t_r}\right)$$

If f is a function of x_is then the uncertainty in f is given by

$$\sigma_f^2 = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i}\sigma_{x_i}\right)^2$$

$$\sigma_{ne}^2 = \left(\frac{Cr^3}{Vt_r}\sigma_{t_f}\right)^2 + \left(\frac{-Cr^3 t_f}{Vt_r^2}\sigma_{t_r}\right)^2$$

The error in times are calculated as

$$\sigma_t = \frac{S_t}{\sqrt{N}}$$

where S_t is the standard deviation and N is the number of readings. Proceeding with the calculation we get the following values.

Value	Error
2.83	0.0313
2.94	0.0353
2.85	0.0157
2.84	0.0340
2.88	0.0158

All values are scaled by 10^{19} to make life easier.

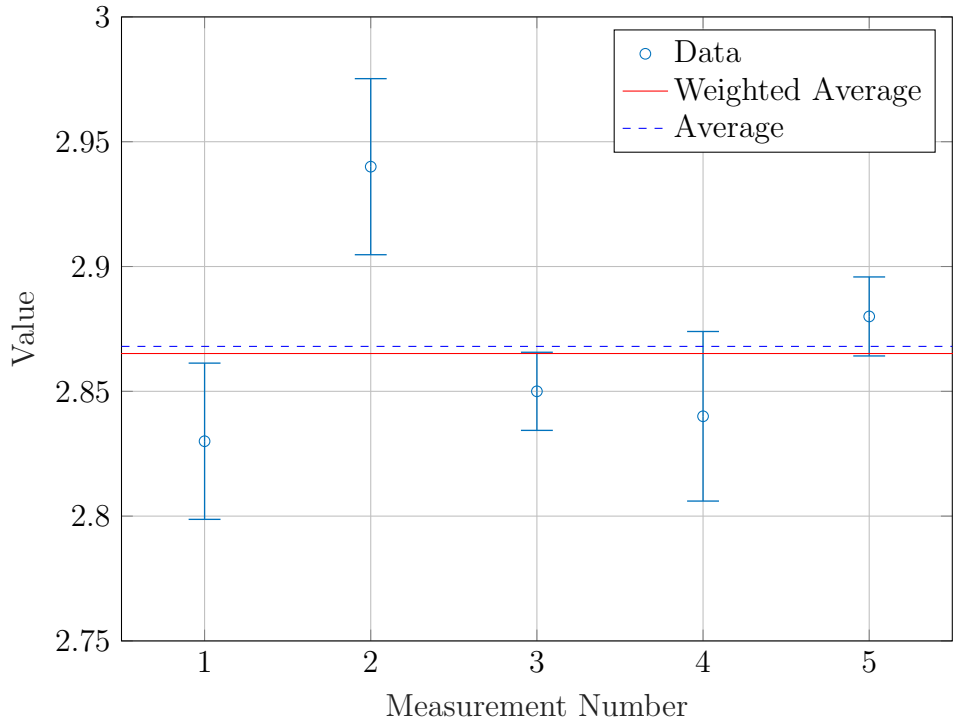


Figure 1: Fitting for weighted average we get the value of charge as 2.865×10^{-19}. Also notice that the average without taking the weights is close to it with the value 2.868×10^{-19}

For balancing methord, the formula from above changes to

$$ne = \frac{Cr^3}{V}$$

This is because T approaches 1 as t_r starts to grow. Thus, even if the oil drop does not achieve rest, as long as it is slow enough we get a good enough approximation for the value of charge. Now,

t_f	v_f	V_b	r	ne
21.43	9.33E-05	405.33	8.66E-07	3.15E-19
19.29	1.04E-04	169.00	9.15E-07	8.92E-19
38.84	5.15E-05	153.00	6.33E-07	3.27E-19
33.48	5.97E-05	221.33	6.85E-07	2.86E-19
11.86	1.69E-04	341.67	1.18E-06	9.40E-19

Table 2: Data calculated for balancing methord

ne/min	n_{eff}	ne/n_{eff}
1.10	1	3.15E-19
3.12	3	2.97E-19
1.14	1	3.27E-19
1.00	1	2.86E-19
3.29	3	3.13E-19

$$\sigma_{ne}^2 = \left(\frac{Cr^3}{V^2}\sigma_V\right)^2$$

Calculating error for these values and plotting the data with error bars, we get Figure 2

Value	Error
3.15	0.0196
2.97	0.0176
3.27	0.1495
2.86	0.0908
3.13	0.0231

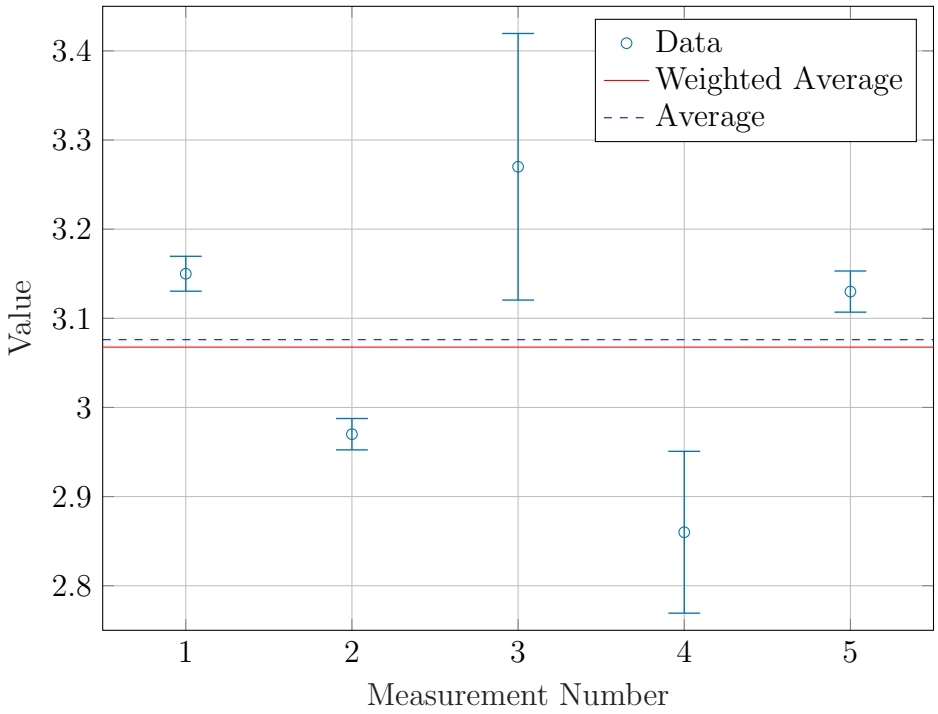


Figure 2: Fitting for weighted average we get the value of charge as 3.068×10^{-19}. Also notice that the average without taking the weights is close to it with the value 3.076×10^{-19}

This gives us the values for charge for both methords as 2.865 and 3.068. Looking at the data, we see that the smallest oil droplet might have contained two electrons. Thus, by

dividing the values by two we get the charge of the electron as

$$1.4325 \times 10^{-19}\ C \text{ for dynamic methord}$$

$$1.534 \times 10^{-19}\ C \text{ for balancing methord}$$

We also notice in the above figures that the actual average and weighted averages are very close to each other. Thus the error can be calculated as

$$\sigma_e = \frac{S_e}{\sqrt{N}}$$

This gives the values of errors to be 0.02×10^{-19} and 0.08×10^{-19} for dynamic and balancing methords respectively.

Result

We have the charge of electron as the following

$$\text{Dynamic methord} : \ 1.43 \pm 0.02 \times 10^{-19}\ C$$
$$\text{Balancing methord} : \ 1.53 \pm 0.08 \times 10^{-19}\ C$$